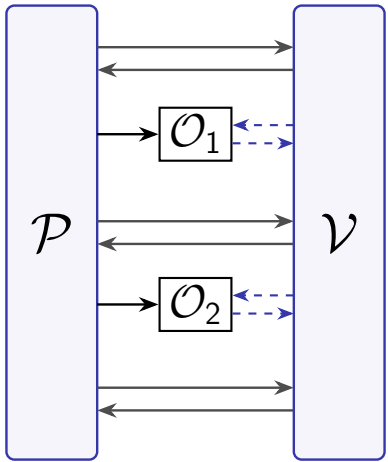
Interactive Oracle Reductions, Revisited

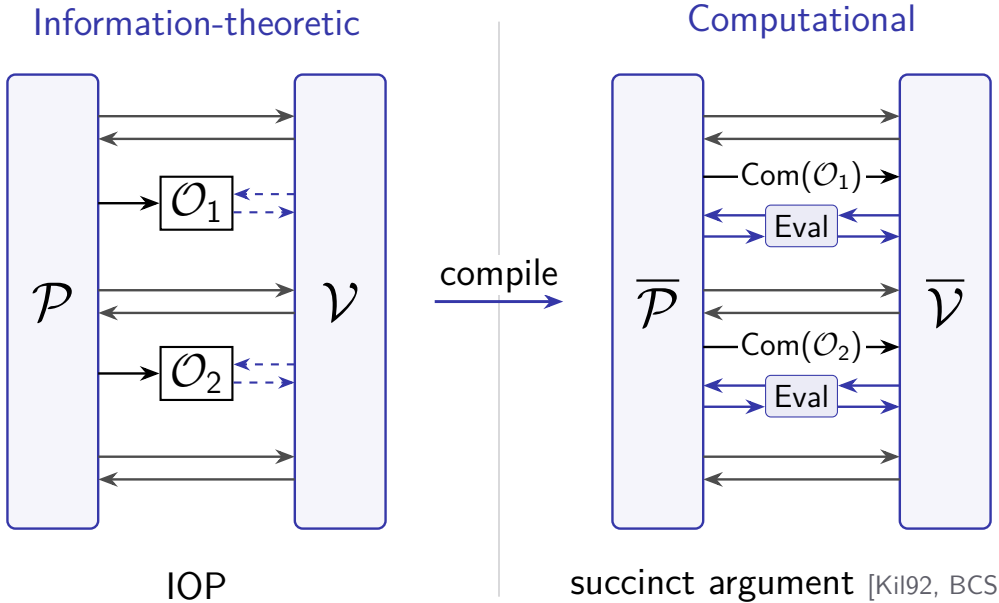
Valerio Cini, Daniele Micciancio, Duc Tu Pham

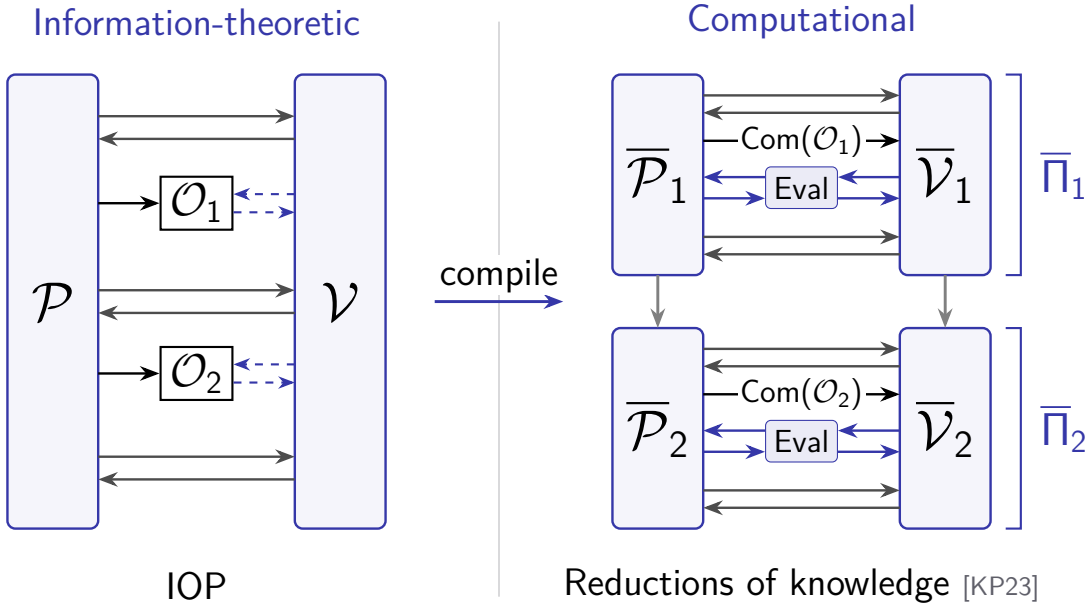
Bocconi UC San Diego ENS | PSL UNIVERSITÉ PARIS

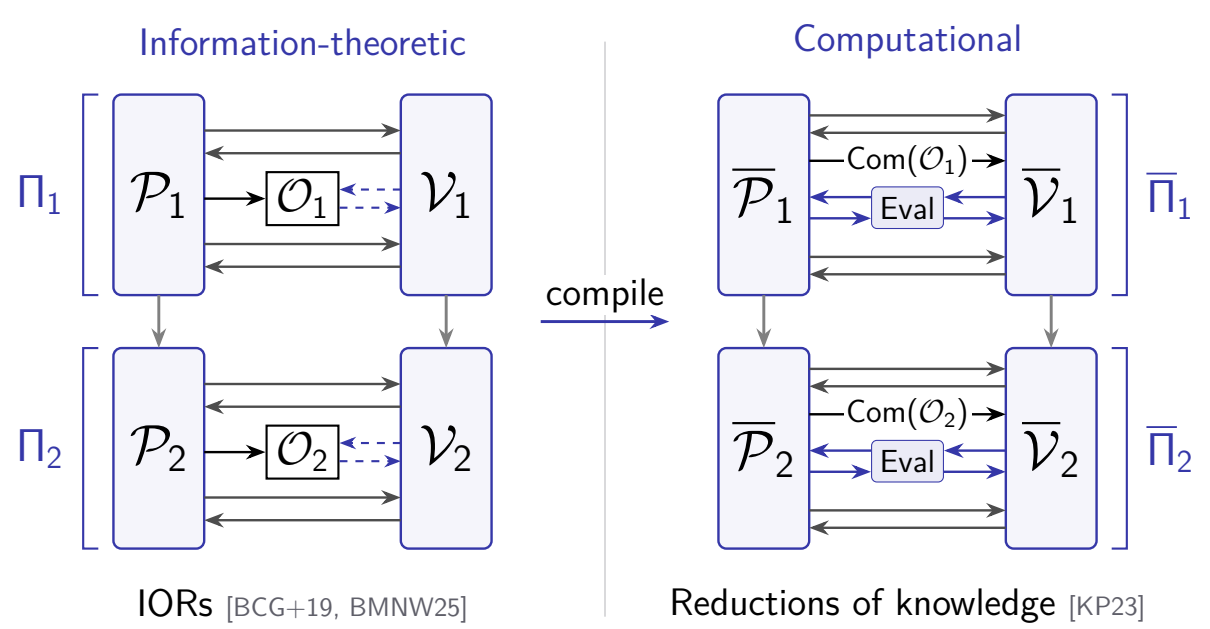
Information-theoretic

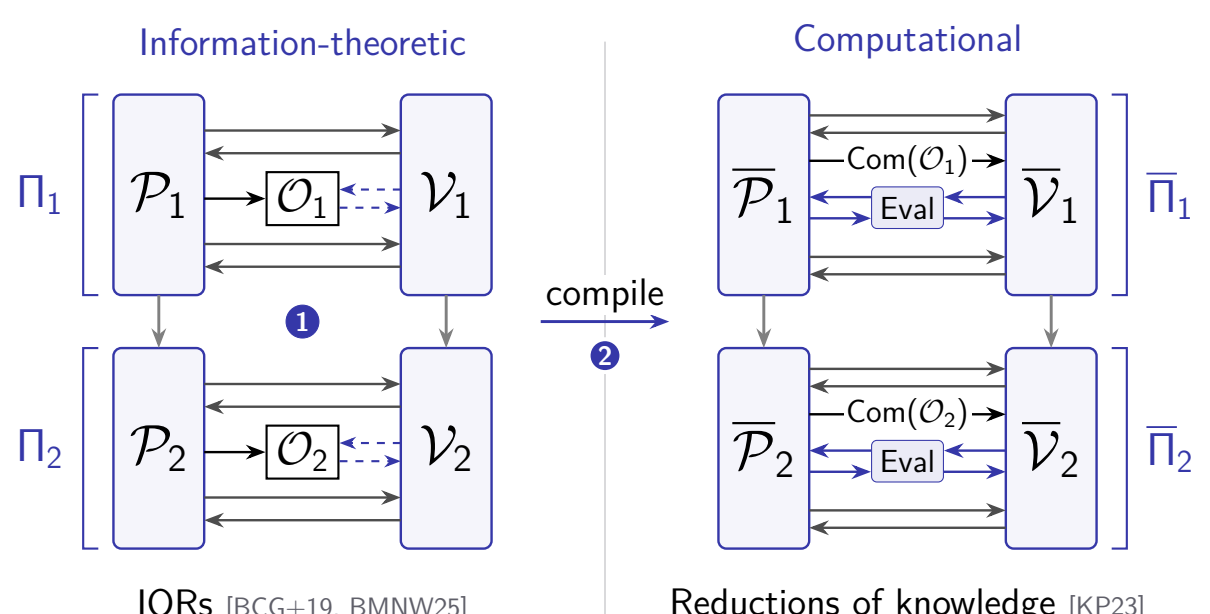


IOP


Information-theoretic
Computational
$\mathcal{P}$
$\mathcal{V}$
$\mathcal{O}_1$
$\mathcal{O}_2$
compile
$\overline{\mathcal{P}}$
$\overline{\mathcal{V}}$
Com($\mathcal{O}_1$)
Eval
Com($\mathcal{O}_2$)
Eval
IOP
succinct argument [Kil92, BCS16]


Information-theoretic
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Π_1
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Information-theoretic
Computational
Π_1
$\mathcal{P}_1$
$\mathcal{O}_1$
$\mathcal{V}_1$
1
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2
$\overline{\Pi}_1$
$\overline{\mathcal{P}}_1$
Com($\mathcal{O}_1$)
Eval
$\overline{\mathcal{V}}_1$
Π_2
$\mathcal{P}_2$
$\mathcal{O}_2$
$\mathcal{V}_2$
$\overline{\Pi}_2$
$\overline{\mathcal{P}}_2$
Com($\mathcal{O}_2$)
Eval
$\overline{\mathcal{V}}_2$
IORs [BCG+19, BMNW25]
Reductions of knowledge [KP23]
1 Do IORs compose?
2 Is compiling them sound?

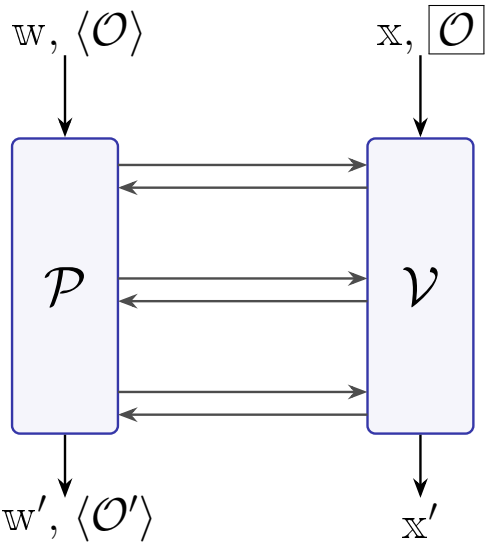
Our contributions

- Definition of IORs (and partial knowledge)
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- Functional commitment with oblivious evaluation

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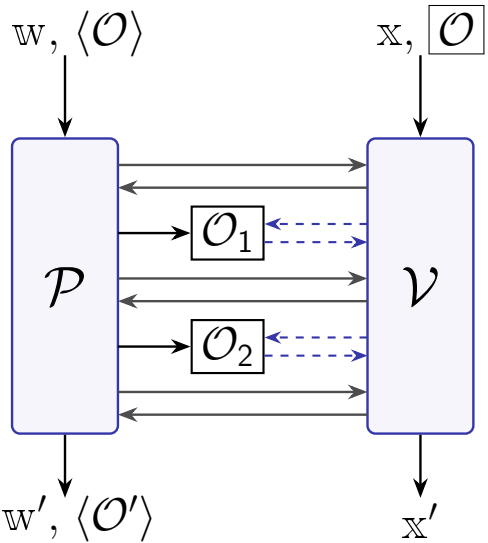
Interactive oracle reductions (IORs)



$$\Pi : \mathsf{R} \longrightarrow \mathsf{R}'$$

$$(\mathbb{x}, \mathbb{w}, \langle\mathcal{O}\rangle) \in \mathsf{R} \;\Rightarrow\; (\mathbb{x}', \mathbb{w}', \langle\mathcal{O}'\rangle) \in \mathsf{R}'$$

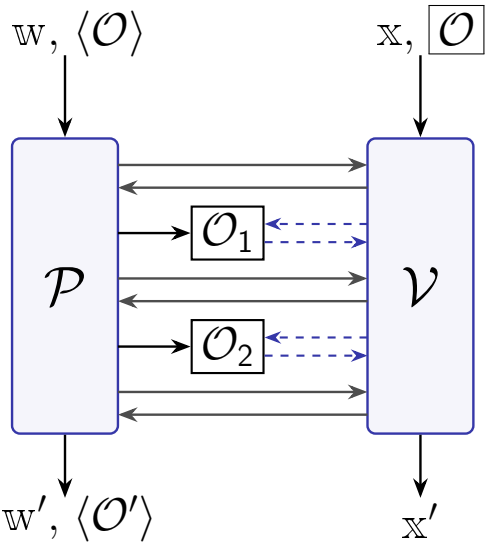
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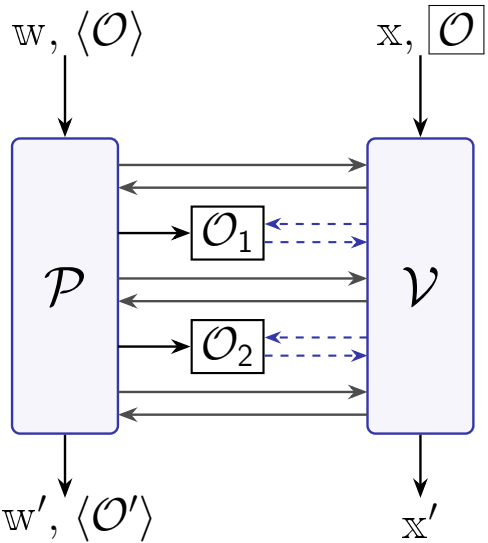
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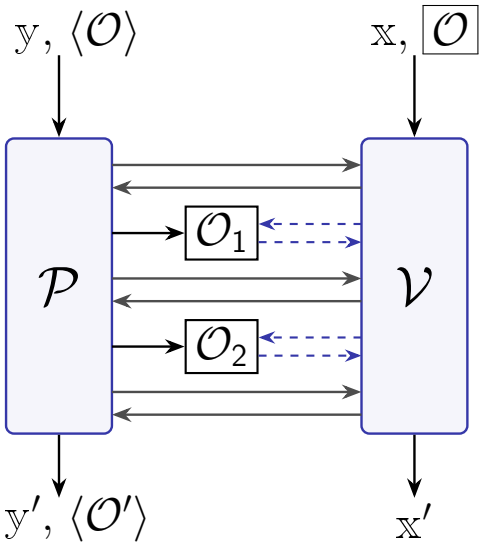
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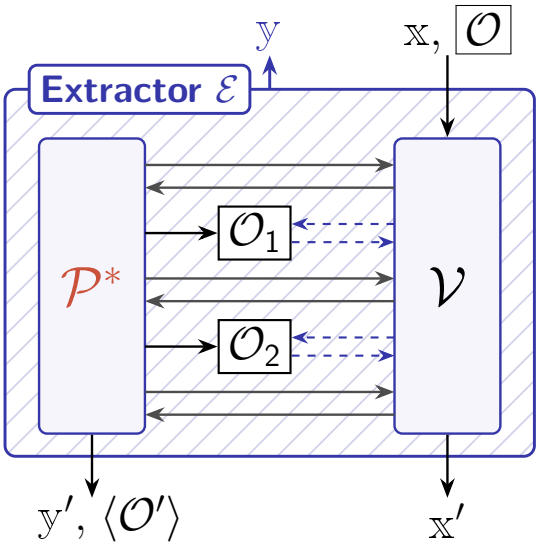
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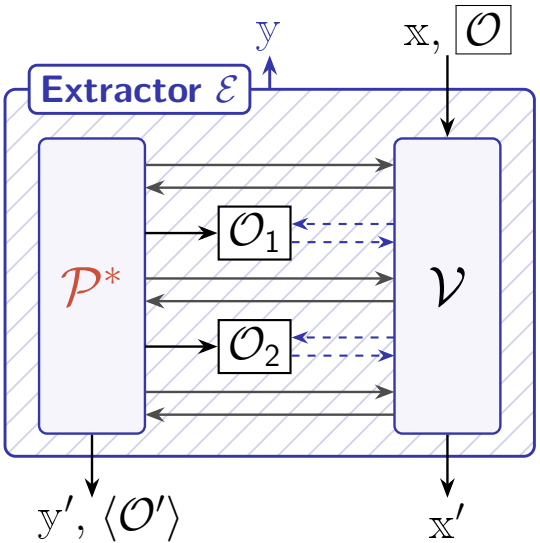
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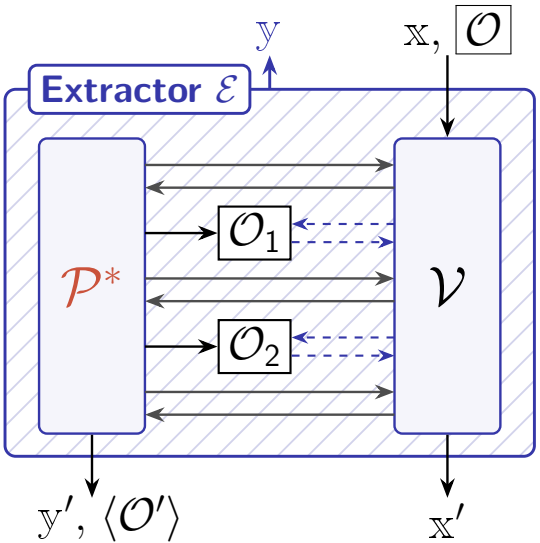
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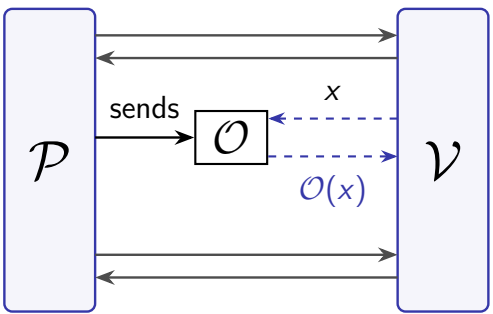
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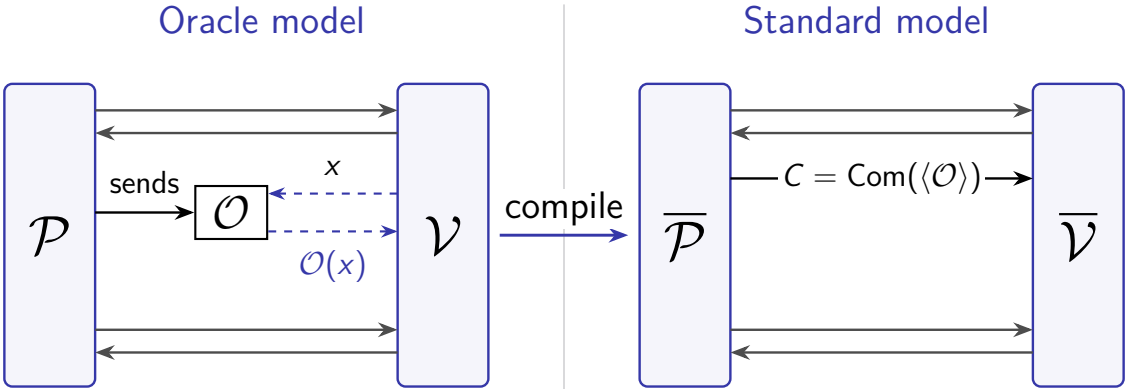
Compiling the oracles away

Oracle model

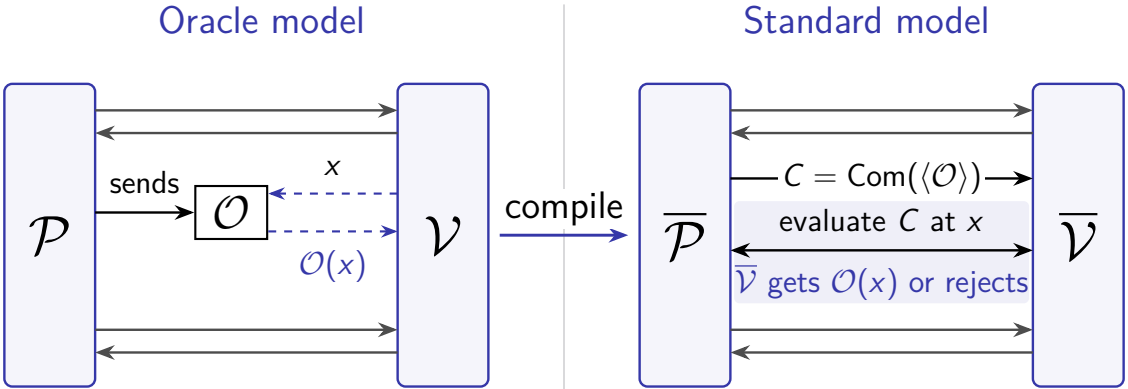


IOR Π

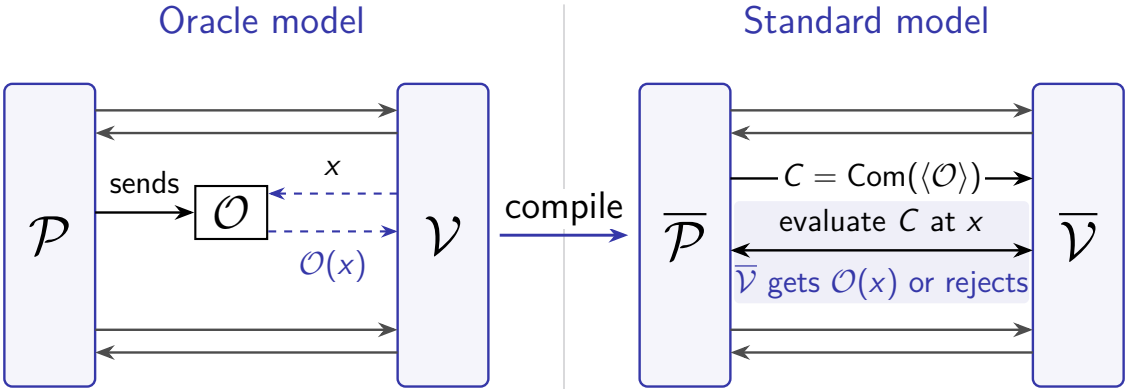
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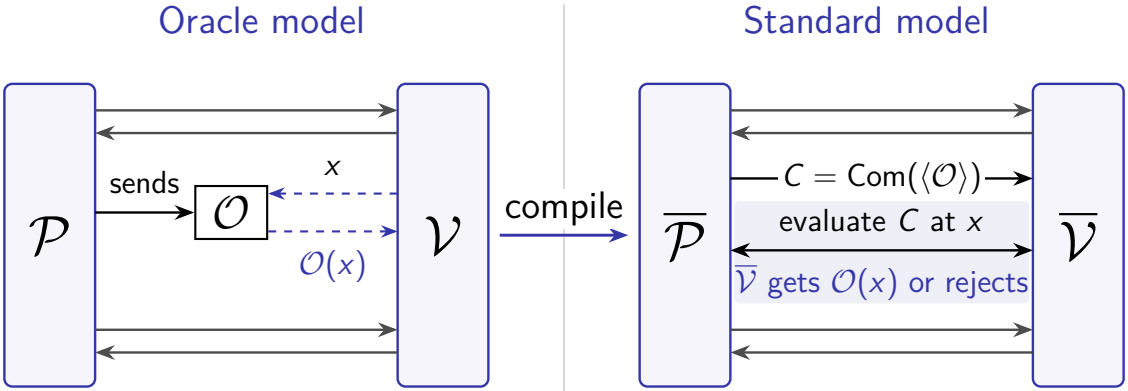


Compiling the oracles away



Π sound $\Longrightarrow$ $\overline{\Pi}$ sound?

Compiling the oracles away



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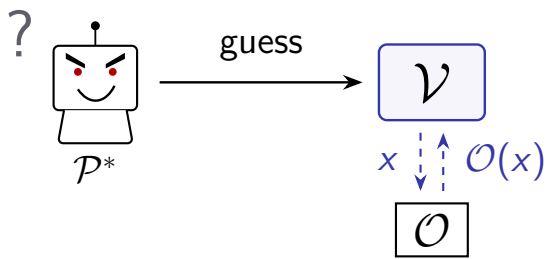
public-coin: ✓

beyond: ?

The evaluation may leak the query

Toy IOR: $\mathcal{V}$ also queries a random x, and accepts if the prover guesses x

Oracle model



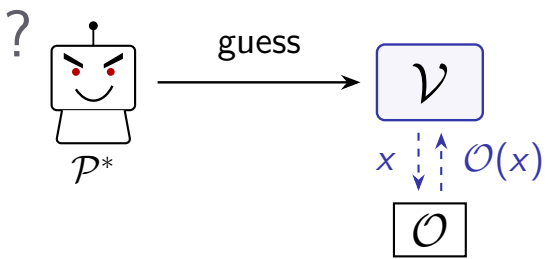
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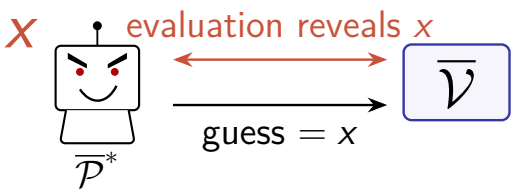
Oracle model



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After compilation



always wins

$\Rightarrow$ not sound

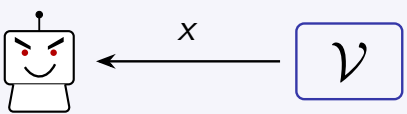
Sound oracle replacement

Theorem. If FC is extractable, then $\overline{\Pi}$ is sound if

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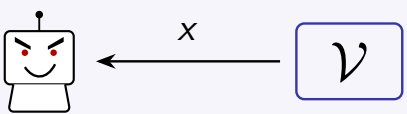


sound even if $\mathcal{P}^*$ learns the queries [CDGS23]

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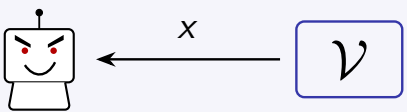


evaluation hides x from $\overline{\mathcal{P}}^*$, and Π is sound for full knowledge

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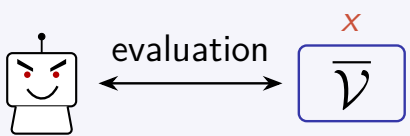
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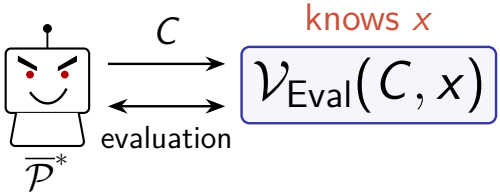


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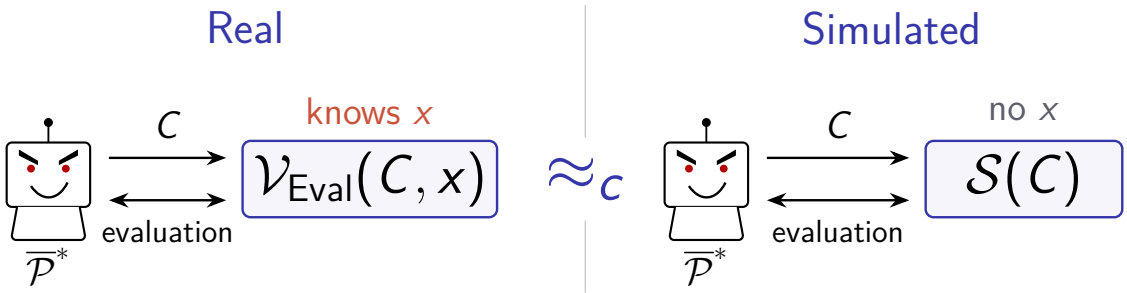
FC construction with oblivious evaluation (SIS + LWE)

Oblivious evaluation

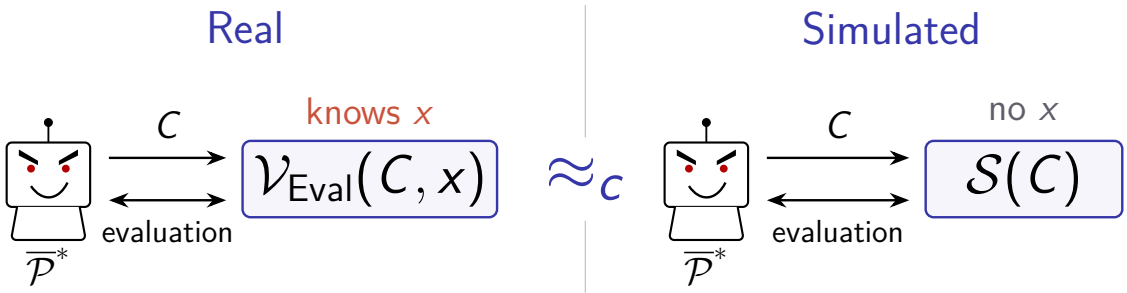
Real



Oblivious evaluation



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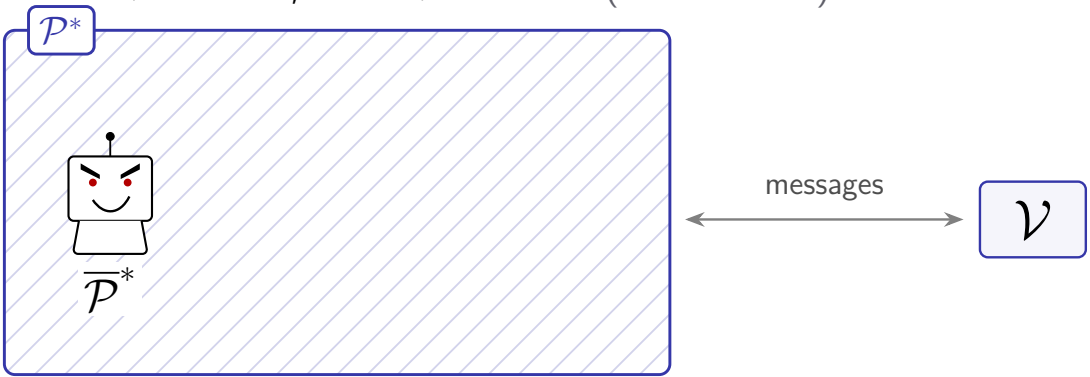


for any committed C: the two views of $\overline{\mathcal{P}}^*$ are indistinguishable

$\Rightarrow$ the evaluation reveals nothing about x

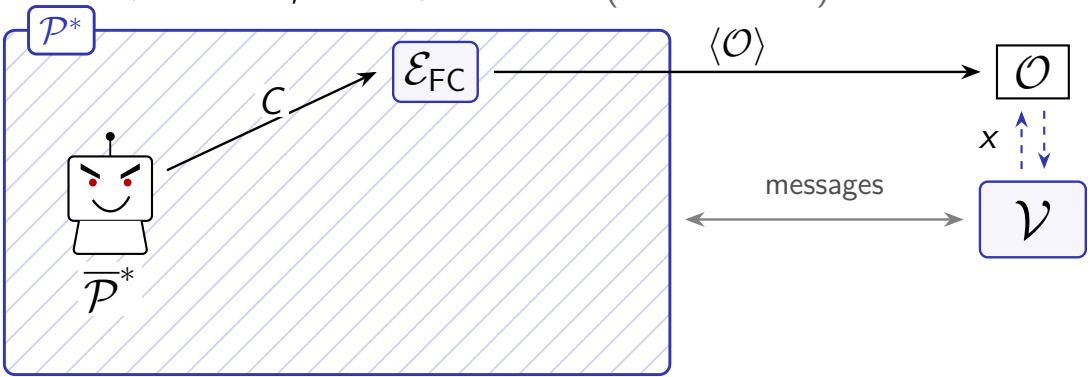
Proof idea

From $\overline{\mathcal{P}}^*$ for $\overline{\Pi}$, build $\mathcal{P}^*$ for Π (oracle model)



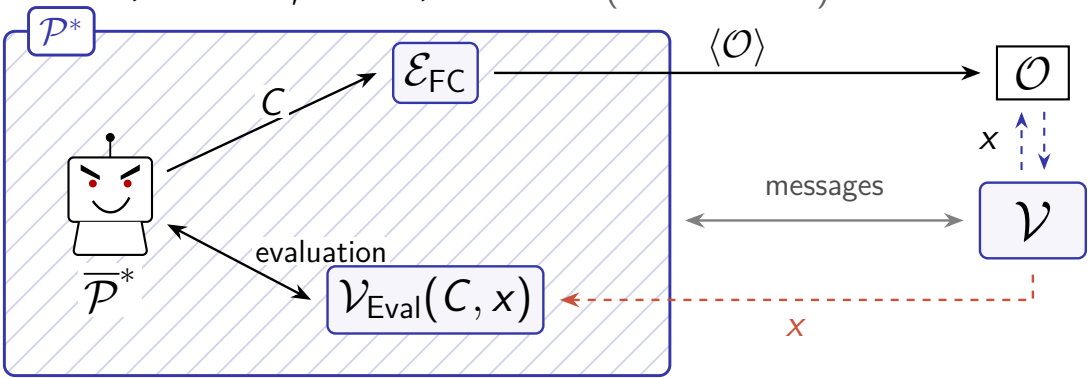
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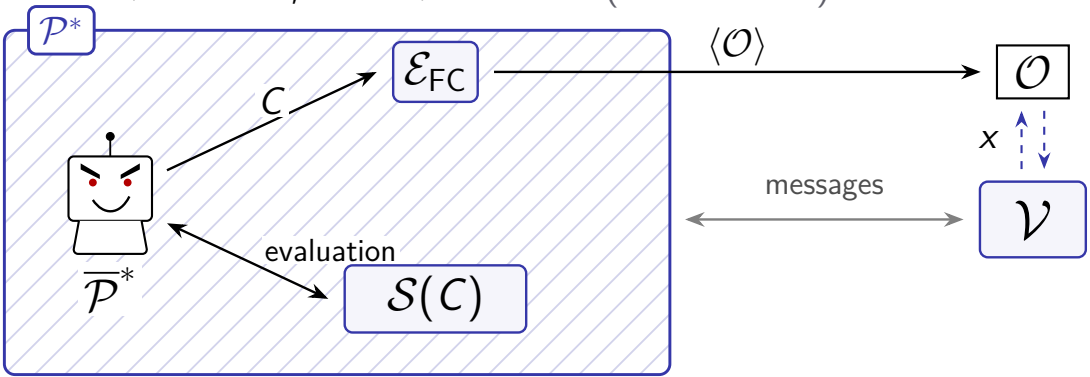
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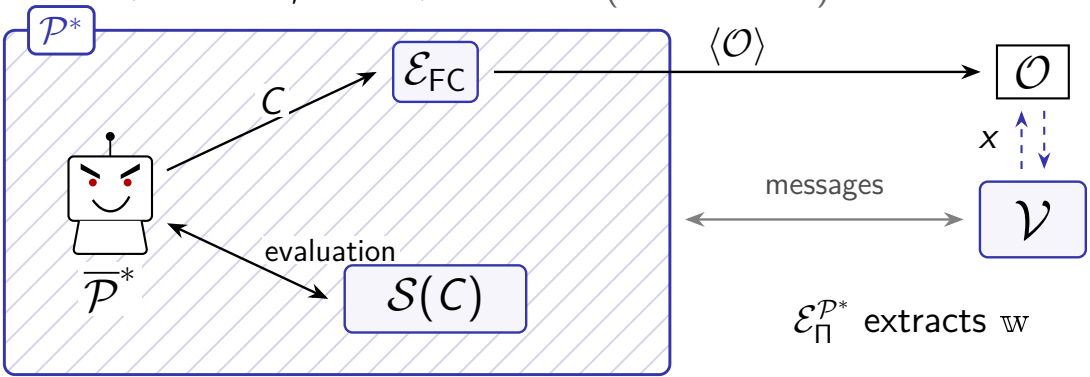


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Thank you!